

Effect of Nb addition on the microstructure and properties of AlCoCrFeNi high-entropy alloy

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ABSTRACT

The microstructures and properties of the AlCoCrFeNb_xNi high-entropy alloys (HEAs) were investigated. Two phases were found in the prepared AlCoCrFeNb_xNi HEAs: one is body-centered-cubic (BCC) solid solution phase; the other is the Laves phase of (CoCr)Nb type. The microstructures of the alloy series vary from hypoeutectic to hypereutectic, and the compressive yield strength and Vickers hardness have an approximately linear increase with increasing Nb content. The residual magnetization (Mr) reaches a maximum for AlCoCrFeNb_{0.1}Ni alloy, which is 6.106 emu/g. The factor of Ω , which is defined as entropy of mixing times 1000 over enthalpy of mixing, well predicts the phase formation for the multicomponents alloys.

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1. Introduction

Generally, the development of new kinds of metallic alloys is ascribed to their excellent properties, e.g., bulk metallic glasses (BMGs) possess high room temperature strength, high elastic strain limits and high corrosion resistance [1,2]; intermetallics possess excellent high-temperature mechanical properties [3]. However, they usually fail catastrophically under loading with very limited macroscopic deformations at room temperature.

However, this impasse has been broken with a startling discovery of so-called high-entropy alloys (HEAs) with multiprincipal elements by using a new alloy design concept of equal/near-equal molar ratios. They are mainly composed of solid solution phases because of their high entropies of mixing. The solid solution phases are beneficial in having a higher work hardening capacity and better plasticity at room temperature as compared to BMGs and intermetallics, which further renders them potential candidates for use as engineering materials [4–7]. Among the alloys investigated, Al_xCoCrFeNi (x value in molar ratio) is an interesting system which has a wide range of microstructures and properties as aluminum content is varied [8–12], and the effects of the secondary elemental additions in this alloy system such as Ti, Si, Mo, and C elements have also been studied in order to further tailor the microstructures and properties of alloys [13–18]. It has been reported that the compressive yield strength, fracture strength and plastic strain limit of

the AlCoCrFeNiTi_{0.5} HEA are 2.26 GPa, 3.14 GPa and 23.3% at room temperature, respectively, which can sustain a high yield strength (~1450 MPa) even at 500 °C due to the Ti-induced secondary-phase strengthening effect [13,19]. Thus in this study, we still selected AlCoCrFeNi alloy as the base alloy and Nb as the alloying element based on the following several reasons: firstly, the Nb element has been successfully used to synthesize the Ti–Al–Nb high temperature/low density intermetallic materials and Zr–Ti–Cu–Be–Nb in situ BMG matrix composites [20,21]; secondly, Nb and other incorporated elements have very negative enthalpies of mixing (the values of Nb and Al, Nb and Co, Nb and Cr, Nb and Fe, Nb and Ni atomic pairs are –18, –25, –7, –16 and –30 kJ/mol, respectively [22]), on the other hand, Nb has the largest atomic size in this alloy system (the relationship of the atomic size r of the incorporated elements is $r_{\text{Nb}} > r_{\text{Al}} > r_{(\text{Cr,Co,Fe,Ni})}$ [23]), both of which maybe favorable to the precipitation out of the secondary phase and the resultant alloy strengthening. Therefore, the AlCoCrFeNb_xNi HEAs ($x = 0, 0.1, 0.25, 0.5$ and 0.75) were synthesized and investigated in detail.

2. Experiment procedures

Ingots with nominal composition of the AlCoCrFeNb_xNi (x value in molar ratio, $x = 0, 0.1, 0.25, 0.5$ and 0.75 , denoted as Nb₀, Nb_{0.1}, Nb_{0.25}, Nb_{0.5} and Nb_{0.75}, respectively) alloys were prepared by arc melting pure elements with purity higher than 99.5 wt% under high-purity argon atmosphere on a water-cooled Cu hearth. The alloys were remelted several times in order to obtain homogeneity. Cylindrical rods with a diameter of 5 mm were prepared by the copper-mould suction casting. Microstructure investigations

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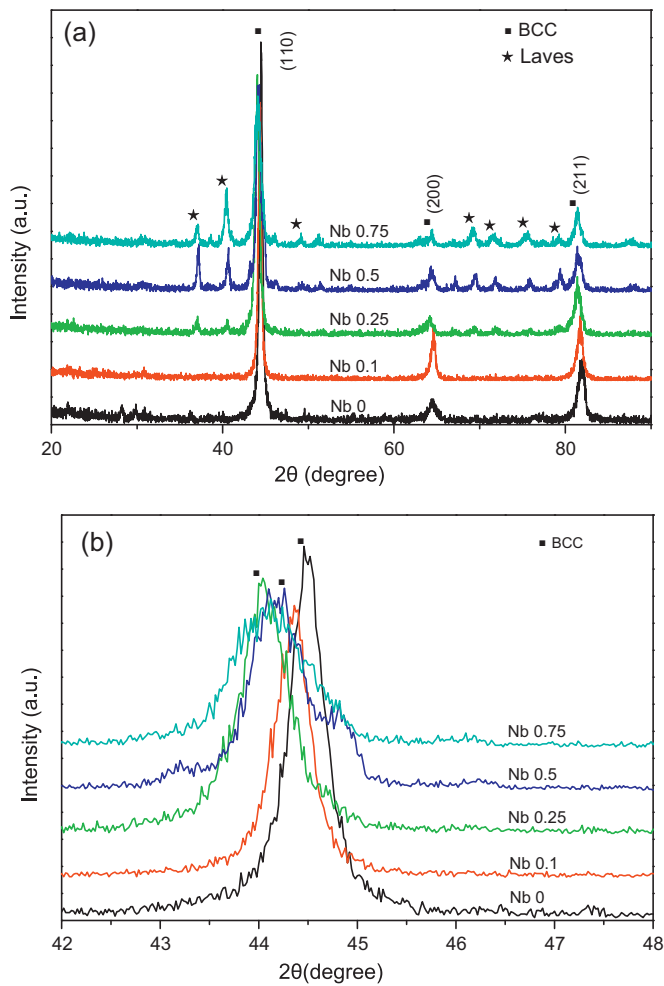


Fig. 1. (a) XRD patterns of the $\text{AlCoCrFeNb}_x\text{Ni}$ ($x=0, 0.1, 0.25, 0.5$, and 0.75) rod samples with a diameter of 5 mm by copper-mould suction casting; (b) the detailed scans for the peak of (1 1 0) of BCC phase.

of the cylindrical alloys were carried out by X-ray diffraction (XRD) using a PHILIPS APD-10 diffractometer with $\text{Cu K}\alpha$ radiation. Cylindrical samples of $\varnothing 5 \text{ mm} \times 10 \text{ mm}$ were prepared for compressive tests and investigated using MTS 809 materials testing machine at room temperature with a strain rate of $2 \times 10^{-4} \text{ s}^{-1}$. The hardness was measured using a Vickers hardness tester under a load of 500 g, holding for 15 s. The morphologies of cross sections and fracture surfaces were examined using a LEO-1540 scanning electron microscope (SEM) with energy dispersive spectrometry (EDS). The magnetization curves were measured by LDJ 9600 vibrating sample magnetometer.

3. Results

3.1. Microstructure evolution

Fig. 1(a) shows the XRD patterns of the as-cast rods for the alloy series. Both AlCoCrFeNi and $\text{AlCoCrFeNb}_{0.1}\text{Ni}$ alloys exhibit the reflections of a single body-centered-cubic (BCC) solid solution phase. However, reflections of the Laves phase can be found in $\text{AlCoCrFeNb}_{0.25}\text{Ni}$, $\text{AlCoCrFeNb}_{0.5}\text{Ni}$ and $\text{AlCoCrFeNb}_{0.75}\text{Ni}$ alloys, which are significantly different from the two formers, and the Laves phase can be identified as (CoCr)Nb type with a hexagonal close-packed (HCP) lattice structure and lattice parameters of $a=0.8601 \text{ nm}$ and $c=0.4733 \text{ nm}$. Furthermore, the relative intensity of the Laves phase is evidently lower than that of the BCC solid

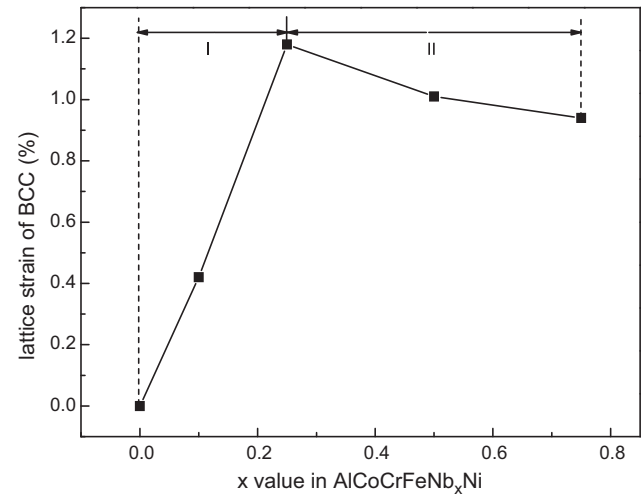


Fig. 2. The lattice strains curve of BCC solid solutions as a function of Nb contents in the $\text{AlCoCrFeNb}_x\text{Ni}$ ($x=0, 0.1, 0.25, 0.5$, and 0.75) alloys.

solution phase, revealing that the main phase still be the latter in this alloy system. This maybe ascribed to the high-entropy effect that greatly lowers the Gibbs free energy of the system [5], which more easily yields the formation of solid solutions during solidification, rather than ordered compounds especially at the high temperature and leads to the total number of phases well below the maximum equilibrium number allowed by the Gibbs phase rule.

Moreover, the detailed scans for the peak of (1 1 0) of the BCC solid solution phase are shown in Fig. 1(b). It is noticed that the peak of (1 1 0) shifts towards lower 2θ as $x < 0.25$ while towards higher 2θ as $x \geq 0.25$. This change is identical to the variation of lattice parameters of BCC solid solutions, which are estimated to be 0.2875, 0.2887, 0.2909, 0.2904, and 0.2902 nm from the strongest (1 1 0) peak. In addition, Fig. 2 shows the lattice strains of BCC solid solutions as a function of Nb contents. The lattice strain can be expressed as $\varepsilon = \Delta a/a_0$, where $\Delta a = |a - a_0|$, here, the crystal of AlCoCrFeNi alloy is assumed as a “perfect” crystal in order to investigate the effect of Nb addition on the lattice strain ε [17]. It is found that the lattice strains first increase (mark I) and then decrease (mark II) with the Nb addition. This indicates that the solution limit of the Nb element into the AlCoCrFeNi solid solution alloy is less than 4.76 at.% corresponding to $x=0.25$, which thus yields the formation of ordered Laves phase.

The microstructures of as-cast samples with a diameter of 5 mm were displayed in Fig. 3. Fig. 3(a) shows a typical dendritic morphology with an average primary arm spacing of 15–25 μm for $\text{AlCoCrFeNb}_{0.1}\text{Ni}$ alloy. Compared to AlCoCrFeNi HEA [13], the main phase is still the BCC solid solution phase corresponding to the DR region, while a small amount of the Laves phase nucleates and grows in the interdendritic regions, as marked by the arrow pointing to the ID location, in spite that the Laves phase was not observed in the XRD patterns because of its small amount. The $\text{AlCoCrFeNb}_{0.25}\text{Ni}$ and $\text{AlCoCrFeNb}_{0.5}\text{Ni}$ alloys (as shown in Fig. 3(b) and (c), respectively) display similar morphologies that exhibit a typical hypoeutectic structure, wherein the primary phase is the BCC solid solution phase (mark A), and the average primary arms spacing of the two alloys are 10–15 μm and 5–10 μm , respectively; the eutectic structure is a mixture of the BCC phase and the Laves phase that nucleate and grow alternately in the interdendritic regions, which constitutes a continuous network structure (mark B). Therefore, it can be seen that the addition of Nb effectively reduces the dendrite size, which may be ascribed to the segregation of Nb element in the interdendrites as shown by EDS analysis results in Table 1. However, for $\text{AlCoCrFeNb}_{0.75}\text{Ni}$ alloy,

Table 1
Chemical composition of as-cast AlCoCrFeNb_xNi (x = 0.1, 0.25, 0.5, and 0.75) alloys.

Alloys	Regions	Al	Co	Cr	Fe	Nb	Ni
AlCoCrFeNb _{0.1} Ni	DR	35.84	15.96	9.29	10.45	3.41	25.05
	ID	12.17	17.20	17.23	17.38	27.65	8.36
AlCoCrFeNb _{0.25} Ni	A	27.22	17.35	15.94	16.35	2.72	20.41
	B	12.20	20.05	22.23	20.30	10.70	14.51
AlCoCrFeNb _{0.5} Ni	A	30.35	17.27	11.39	12.57	2.92	25.49
	B	16.60	18.26	18.41	17.60	13.37	15.75
AlCoCrFeAlNb _{0.75} Ni	A	9.22	17.76	19.51	18.45	25.72	9.33
	B	26.92	16.78	12.60	13.35	7.45	22.90

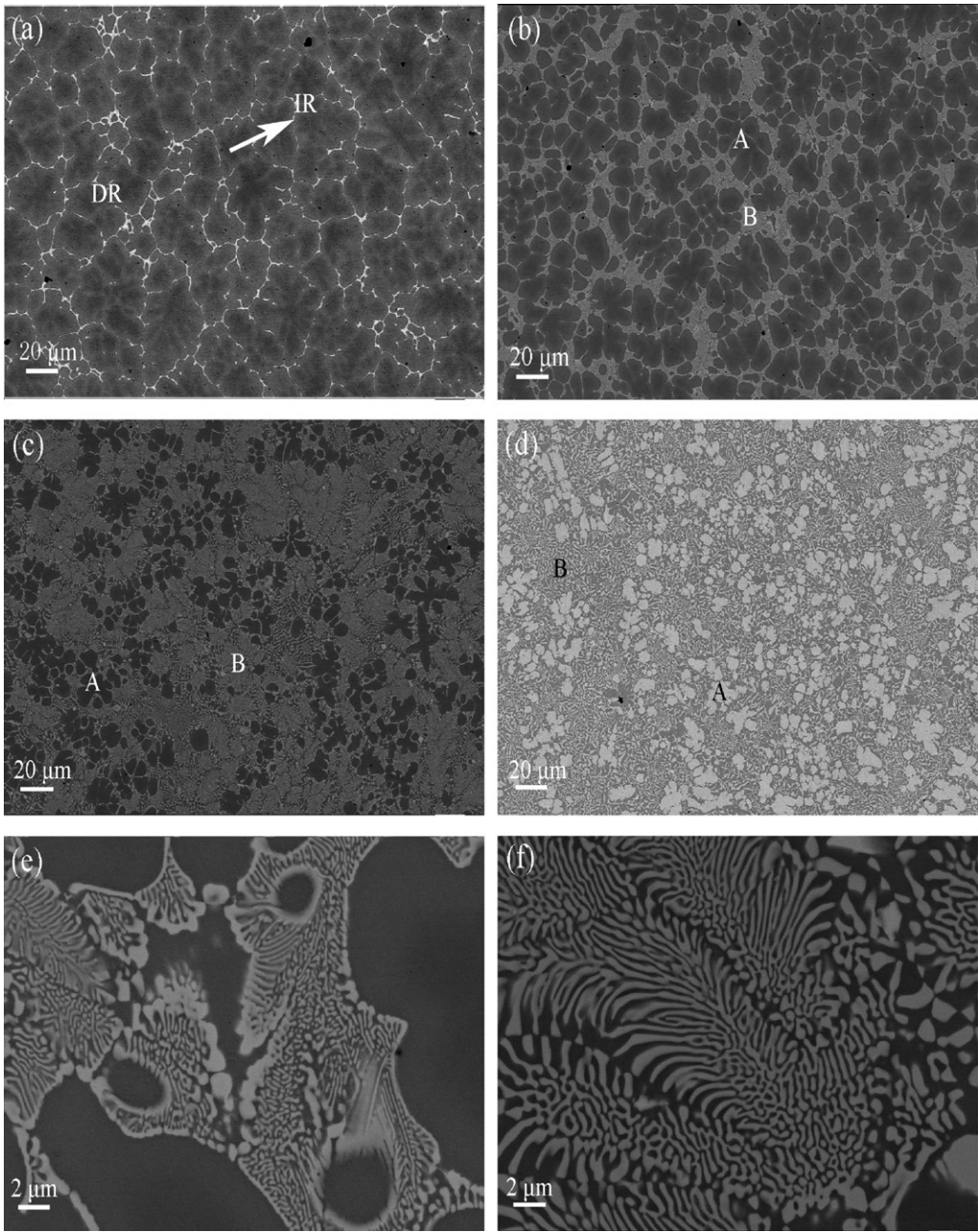


Fig. 3. SEM back-scatter electron images of the AlCoCrFeNb_xNi alloys, x = 0.1, 0.25, 0.5, and 0.75 corresponding to (a), (b), (c) and (d), respectively; (e) and (f) are the magnification of B region in (b) and (c), respectively.

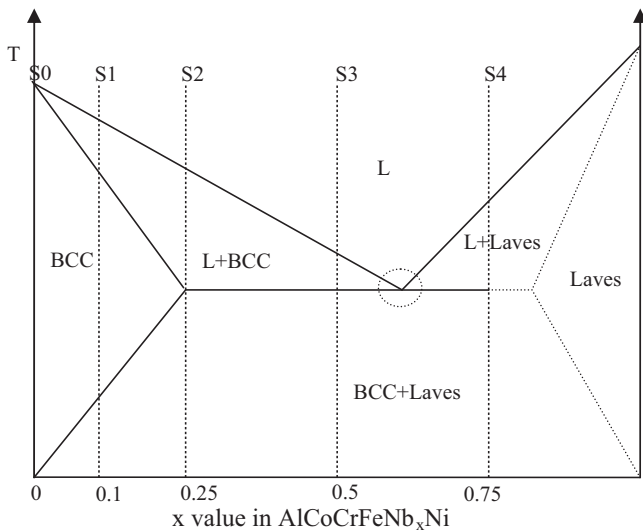


Fig. 4. A schematic phase diagram of the AlCoCrFeNb_xNi alloys, the S0, S1, S2, S3 and S4 corresponding to $x = 0, 0.1, 0.25, 0.5$, and 0.75 , respectively.

the primary phase is not the BCC phase but the Laves phase of a white flower-shaped morphology (mark A) in Fig. 3(d) and the eutectic structure consists of the BCC phase and the Laves phase that nucleate and grow alternately around the crystal boundaries of the Laves phase (mark B), which thus presents a hypereutectic structure for the alloy. As a result, the microstructure changes from single BCC solid solutions ($x = 0$ and $x = 0.1$) to a hypoeutectic structure ($x = 0.25$ and $x = 0.5$) and then to a hypereutectic structure ($x = 0.75$) in this alloy system, of which a schematic phase diagram was subsequently plotted in Fig. 4, revealing that the eutectic point maybe at between $x = 0.5$ and $x = 0.75$ and the eutectic reaction can be expressed as $L \rightarrow \text{BCC} + \text{Laves}$. This reaction may be due to Nb and other transition elements having the preference of eutectic reactions according to the binary phase diagrams [24]. Fig. 3(e) and (f) are the magnification of the eutectic structures in Fig. 3(b) and (c), respectively. It is obvious that the eutectic lamellar spacing λ for AlCoCrFeNb_{0.5}Ni alloy ($\lambda \approx 500$ nm) is larger than that of AlCoCrFeNb_{0.25}Ni alloy ($\lambda \approx 300$ nm). As λ is inverse to the undercooling in the solid/liquid interface front during solidification, which can be expressed as $\lambda \propto (1/\Delta T)$ [25], thus the addition of Nb may play a role in reducing the degree of undercooling in the solid/liquid interface front.

In addition, the EDS results show that the BCC solid solution phase is enriched in Al and Ni and depleted in Nb, while Al and Ni depleted and Nb enriched for the Laves phase as shown in Table 1. This may be ascribed to several reasons: Firstly, Ni has a relatively small atomic size, thus it is easily dissolved into the supersaturated BCC solid solutions during solidification; Al and Ni have similar face-centered-cubic (FCC) lattice structures as well as a large bonding energy because of a very negative enthalpy of mixing (the value of Al and Ni atomic pair is -22 kJ/mol), both of which facilitate the substitution of each other in the lattice site and segregation in the BCC phase matrix; Secondly, as mentioned above, Nb has the largest atomic size in this alloy system as well as very negative enthalpies of mixing with other alloying elements, which are not favorable to the solution of Nb into the BCC solid solutions and conversely enhance the tendency to form the ordered Nb-rich Laves phase.

3.2. Mechanical properties

Fig. 5 shows the compressive engineering stress–strain curves for the alloy series (not including AlCoCrFeNb_{0.75}Ni alloy). The yield strength, fracture strength and plastic strain limits were listed in

Table 2

Mechanical properties and Vickers hardness of as-cast AlCoCrFeNb_xNi ($x = 0, 0.1, 0.25$, and 0.5) alloys.

x	$\sigma_{0.2}$ (MPa)	$\sigma_{\max}$ (MPa)	ε_p (%)	HV
0	1373	3531	24.5	520 ± 11
0.1	1641	3285	17.2	569 ± 11
0.25	1959	3008	10.5	668 ± 12
0.5	2473	3170	4.1	747 ± 10

$\sigma_{0.2}$, yield strength; $\sigma_{\max}$, fracture strength; ε_p , plastic strain limits.

Table 2. It can be seen that the yield strength has a substantial increase from 1373 MPa to 2473 MPa while the plastic strain limits significantly decreases from 24.5% to 4.1% with the Nb addition. The Vickers hardness as a function of Nb contents was plotted in Fig. 6 and the corresponding values were also listed in Table 2. It is found that the Vickers hardness has an approximately linear increase with increasing Nb content, which can be roughly expressed as $y_{HV} = 454x + 530$, where y_{HV} is the Vickers hardness, and x is the Nb content.

The fractographs of the deformed samples after compressive tests were observed in Fig. 7. Fig. 7(a) and (b) shows similar fracture morphologies of a radial herringbone fracture for AlCoCrFeNb_{0.1}Ni and AlCoCrFeNb_{0.25}Ni alloys, and the fracture pattern is of quasi-cleavage. A visible cleavage fracture pattern of the mixture of

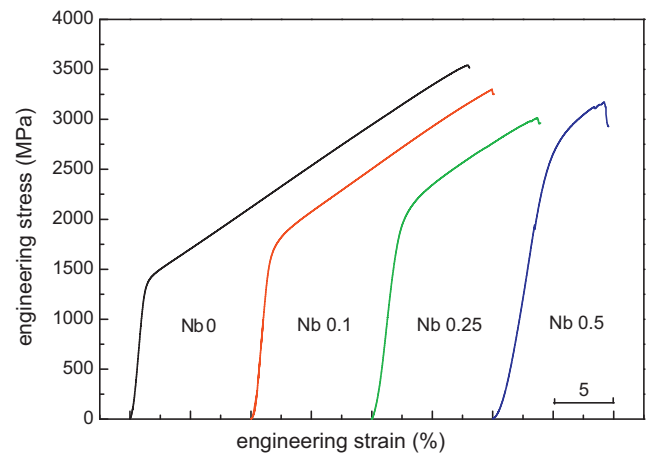


Fig. 5. The compressive stress–strain curves of the AlCoCrFeNb_xNi rod samples with a diameter of 5 mm ($x = 0, 0.1, 0.25$, and 0.5).

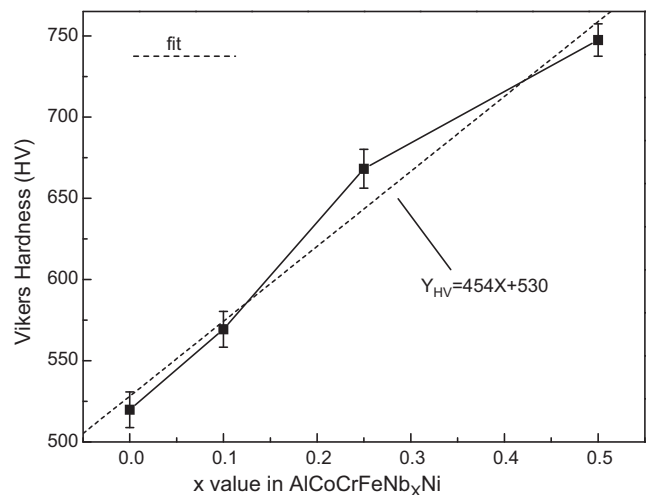


Fig. 6. The Vickers hardness curve of the AlCoCrFeNb_xNi alloys as a function of Nb contents ($x = 0, 0.1, 0.25$, and 0.5).

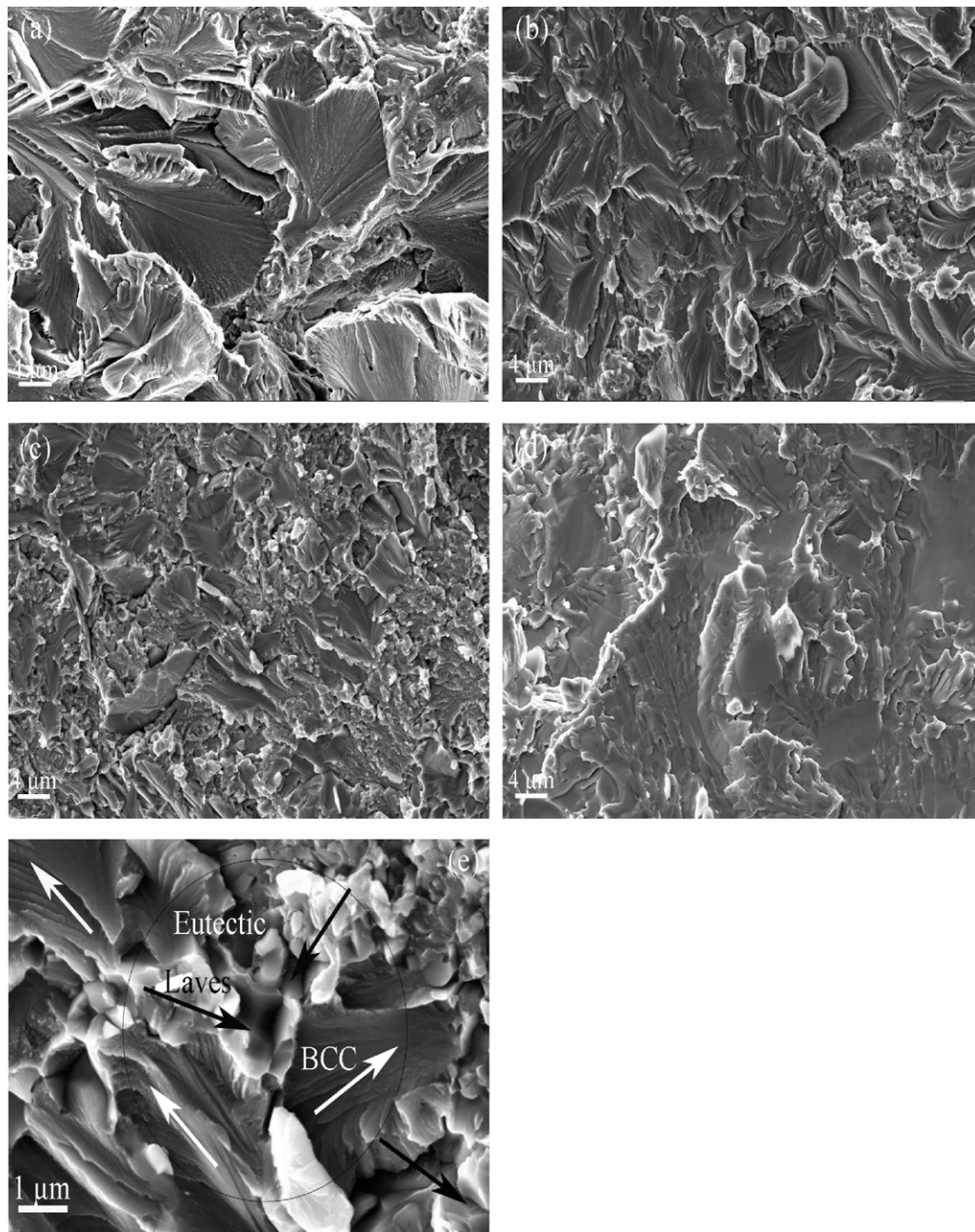


Fig. 7. SEM secondary electron images of fracture surfaces after compressive tests for the AlCoCrFeNb_xNi alloys, $x = 0.1$ (a), 0.25 (b), 0.5 (c) and (e), and 0.75 (d).

inter-granular (dominant) and trans-granular (subordinate) modes can be seen in Fig. 7(c), which thus deteriorates the alloy plasticity. Fig. 7(d) shows a fracture surface with a bright and flat fracture, which accords with the completely brittle fracture of the alloy. The detailed morphology of the eutectic structure composed of BCC phase and Laves phase in the interdendritic region for AlCoCrFeNb_{0.5}Ni alloy is shown in Fig. 7(e), wherein the BCC phase was marked by the white arrow and the Laves phase by the black arrow. One can see that a series of cleavage steps are present in the BCC phase region, while a fragile and block-stacking fracture that seems like blooming flowers can be seen in the Laves phase region, which confirms the cleavage fracture mode.

Moreover, the volume fractions of the Laves phase in this alloy system can be defined as $f_L = \text{Laves}/(\text{BCC} + \text{Laves})$, whose curve as a function of Nb contents was plotted in Fig. 8 and the corresponding values were listed in Table 3. It can be seen that the f_L increases from

0% to 48.69% with increasing Nb content. Thus the mass distribution of the Laves phase along the interdendrites can effectively increase the resistance to the dislocation movement, which is believed to have important contributions to the yield strength and hardness of alloys. Besides, the dendrite size refinement has also a positive contribution. However, the plasticity of the alloys is obviously deteriorated as a result of the increase of the brittle Laves phase as well as the larger eutectic lamellar spacing as the Nb content increases [26].

3.3. Magnetic properties

The magnetization curves of the alloy series are obtained in Fig. 9(a). Clearly, the saturation magnetizations of the AlCoCrFeNb_xNi alloys are 64.71, 49.56, 34.69, 17.61, 10.31 emu/g,

Table 3

Enthalpies of mixing, entropies of mixing, Ω , atomic size difference, phases and volume fractions of the Laves phase for the AlCoCrFeNb_xNi ($x = 0, 0.1, 0.25, 0.5$, and 0.75) alloys.

Alloys	ΔH_{mix} (kJ/mol)	ΔS_{mix} (J/kmol)	Ω	δ (%)	Phase(s)	f_L (%)
AlCoCrFeNi	-12.32	13.38	1.09	5.32	BCC	0
AlCoCrFeNb _{0.1} Ni	-13.32	13.88	1.04	5.62	BCC	4.25
AlCoCrFeNb _{0.25} Ni	-14.66	14.29	0.98	5.99	BCC + (Laves + bcc)	21.10
AlCoCrFeNb _{0.5} Ni	-16.58	14.71	0.89	6.46	BCC + (Laves + bcc)	41.55
AlCoCrFeNb _{0.75} Ni	-18.03	14.87	0.83	6.81	Laves + (Laves + bcc)	48.69

ΔH_{mix} , the enthalpy of mixing; ΔS_{mix} , the entropy of mixing; Ω , a factor reflecting the competitive relationship between ΔH_{mix} and ΔS_{mix} ; δ , atomic size difference; f_L , volume fractions of the Laves phase.

respectively, under 20,000 Oe magnetic field. According to the following equation of the classification of magnetic properties [27]:

$$\chi = \frac{M(A/m)}{H(A/m)} = \frac{\sigma \text{ (emu/g)} \cdot \rho \text{ (g/cm}^3\text{)}}{4\pi \cdot H \text{ (Oe)}} \quad (1)$$

where χ is the permeability, M is the magnetization, H is the magnetic field intensity, σ is the specific saturation magnetization, ρ is the density of alloys and roughly $6.5\text{--}7.5\text{ g/cm}^3$ for the alloy series. According to Eq. (1), the χ can be estimated to be $2.0 \times 10^{-2}\text{--}3.0 \times 10^{-3}$, so it can be seen that this alloy system exhibits a ferromagnetic property, which may be ascribed to the addition of the paramagnetic elements (Al and Nb) and the anti-ferromagnetic element (Cr) into these alloys besides the ferromagnetic elements (Fe, Co and Ni) that do not enable to preserve the ferromagnetic property. Furthermore, the corresponding saturation magnetizations M_s , residual magnetizations M_r and coercive forces H_c are displayed in Fig. 9(b). Based on this, one can see that H_c ranges from 52 to 94 Oe (that is $4.1 \times 10^3\text{--}7.5 \times 10^3\text{ A/m}$), which reveals its soft magnetic behavior; M_s and M_r approximately decrease while H_c increases with increasing Nb content, which may be attributed to two reasons: Firstly, the CoCrNb alloy belongs to hardening magnetic materials and the H_c can reach up to 10^5 A/m [28]. Secondly, the Laves phase may pin the magnetic domain of BCC phase matrix as they are turning under an external magnetic field.

4. Discussion

According to the Hume–Ruthery rule: to comprise the substitution solid solution, the atomic size difference between two incorporated elements should not be larger than 15%, the lattice structure is similar to each other, and the electronegativity has no

big difference. Accordingly, for the alloy series, besides the high entropy of mixing, the atomic size difference and the enthalpy of mixing of the incorporated elements cannot also be ignored. This work was mainly according to the following equations [19,29–31] and the corresponding results were listed in Table 3.

$$\delta = \sqrt{\sum_{i=1}^n C_i \left(1 - r_i / \sum_{i=1}^n C_i r_i \right)^2} \quad (2)$$

where δ is the atomic size difference; C_i is the molar fraction of i th element; r_i is the atomic size of i th element.

$$\Omega = \frac{T \Delta S_{mix}}{|\Delta H_{mix}|} \quad (3)$$

where T is the absolute temperature and assumed as 1000 K; ΔS_{mix} is the entropy of mixing [5]; ΔH_{mix} is the enthalpy of mixing [29–31]; Ω is a factor that is defined as entropy of mixing times

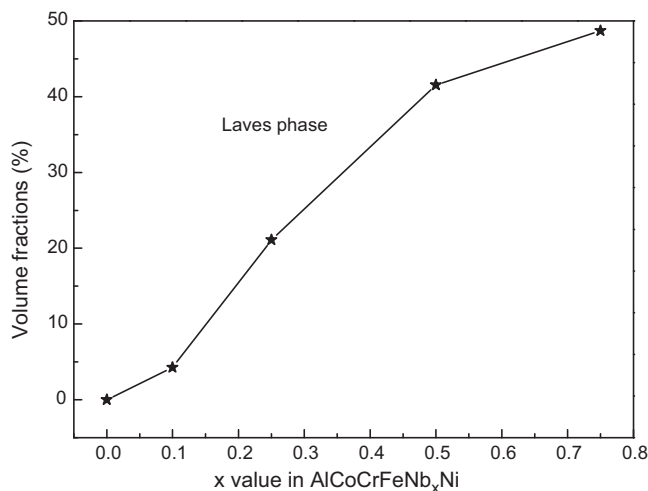


Fig. 8. The volume fractions curve of Laves phase as a function of Nb contents for the AlCoCrFeNb_xNi alloys ($x = 0, 0.1, 0.25, 0.5$, and 0.75).

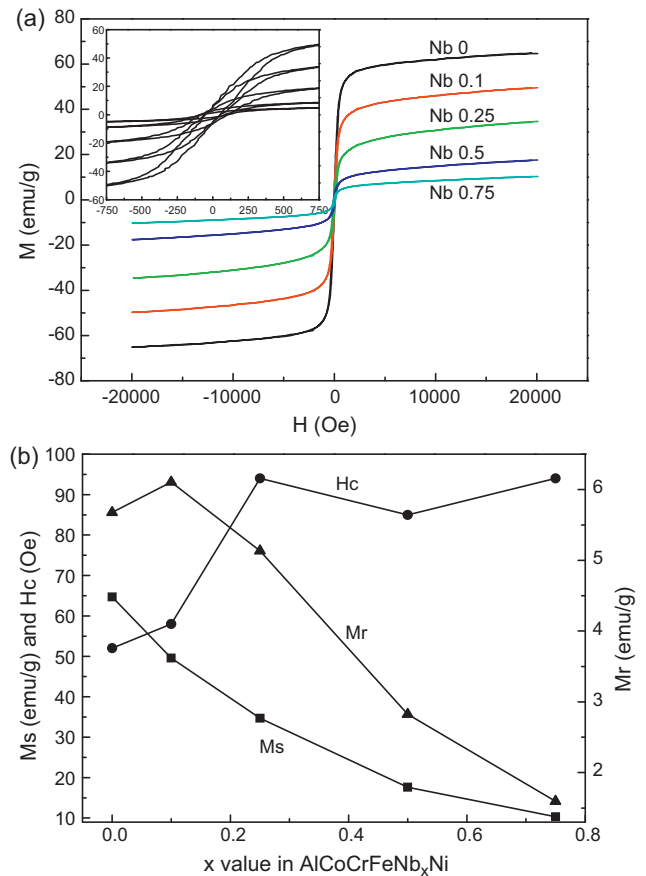


Fig. 9. (a) Magnetization curves of the AlCoCrFeNb_xNi alloys ($x = 0, 0.1, 0.25, 0.5$, and 0.75); (b) presents the corresponding saturation magnetizations M_s , residual magnetizations M_r , and coercive forces H_c .

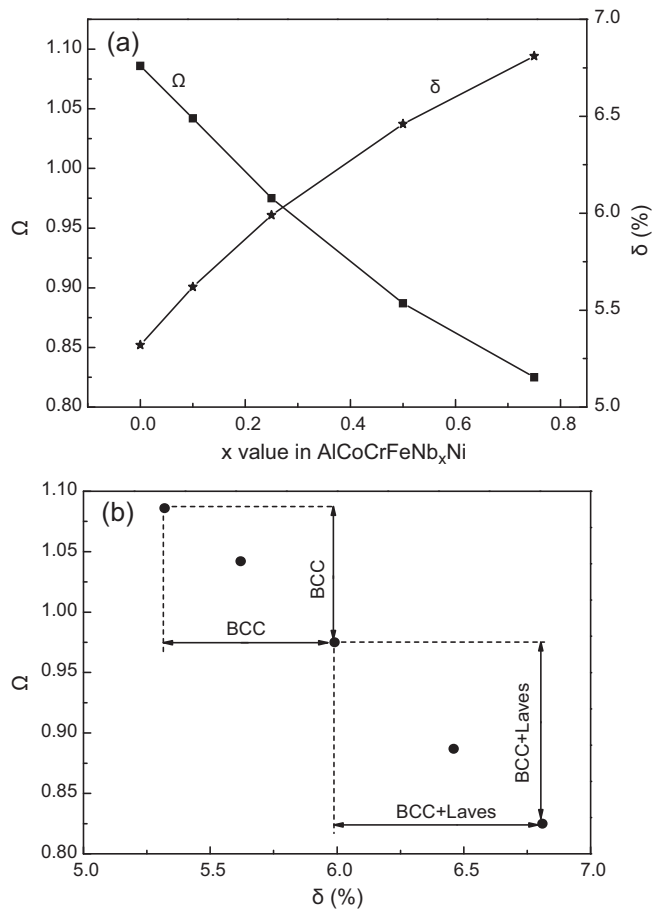


Fig. 10. (a) The curves of Ω and δ as a function of Nb contents for the AlCoCrFeNb_xNi alloys ($x = 0, 0.1, 0.25, 0.5$, and 0.75); (b) gives the correlation between Ω and δ .

1000 over enthalpy of mixing, reflecting the competitive relationship between ΔS_{mix} and ΔH_{mix} .

Subsequently, the curves of Ω and δ as a function of Nb contents were plotted in Fig. 10(a). Clearly, the addition of Nb atoms into the alloys leads to larger atomic size differences from 5.32% to 6.81% and the resultant increase of lattice distortion energy of BCC solid solutions; the Ω values decrease from 1.09 to 0.83 with the Nb addition, which indicates that the entropy of mixing plays a negative role compared to the enthalpy of mixing in this alloy system. As a consequence, the volume fractions of the ordered Laves phase increase and the plasticity of alloys resultantly decreases as the Nb content increases. Therefore, it can be concluded that the larger the Ω value and the smaller the δ value, the more the amount of BCC solid solutions and the less the amount of the Laves phase, which accords with the column of phases in Table 3. Moreover, Fig. 10(b) gives the correlation between Ω and δ of the alloy series. It can be seen that the BCC phase belongs to the region of $0.98 < \Omega < 1.09$ and $5.99 > \delta > 5.32$, while the BCC and Laves phases belong to another region of $0.83 < \Omega < 0.98$ and $6.81 > \delta > 5.99$. Based on this, it may reveal that the region of $\Omega > 1.09$ and $\delta < 5.32$ will tend to form

the solid solution phase, while the region of $\Omega < 0.83$ and $\delta > 6.81$ tending to form the Laves phase.

5. Conclusions

In conclusion, the addition of Nb element into AlCoCrFeNi HEA changes the original phase constitution, which yields the formation of ordered Laves phase besides solid solution phase. This induces a changing microstructure from hypoeutectic to hypereutectic and dual effects on the properties of alloys with different Nb contents, wherein the yield strength, hardness, and coercive forces are increased while the plasticity, saturation magnetizations, and residual magnetizations decreased. Furthermore, the dependence of the microstructures of the alloys on the Nb contents was demonstrated by a schematic phase diagram, and by the introduction of the factors of Ω and δ , the phase formation for the multicomponents alloys was well predicted.

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